

Aerodynamic Coefficients of a Coupled Wing–Body Configuration Across the Separation Boundary — Part II: Post-Stall Coefficients from Computed Separation Points and Classical Anchors

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Part I of this series bounded the attached-flow coefficient tables of a coupled wing–body panel solve at the separation boundary its own boundary-layer march draws ($\alpha \approx 12^\circ$ on the studied tail-sitter configuration). This paper reports what lies beyond, out to $\alpha = 90^\circ$ and $\beta = 30^\circ$ — the regime a tail-sitter transits by definition. The vehicle is a sectional-feedback strip coupling on the same lattice, with a post-stall section model whose every ingredient is either classical theory with published validation or the solve’s own computed separation state: Kirchhoff’s free-streamline attenuation $((1 + \sqrt{f})/2)^2$ applied to the exact thin-airfoil force pair, with the separation point f interpolated from Part I’s marches; the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling $C_{d,\max} = 1.11 + 0.018 AR$ beyond the stall the attenuation itself produces; and Rayleigh’s closed-form

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centre of pressure for the section pitching moment. Stall is thereby an output — $C_{L,\max} = 1.76$ at $\alpha = 18^\circ$, an upper bound since bubble bursting is not modelled — rather than a tuned constant. Across the map, sideslip collapses onto the total inclination $\sin \sigma = \sin \alpha \cos \beta$ to within 13%, the flat-plate regime is reached by $\alpha \approx 55\text{--}65^\circ$, the computed $C_N(90^\circ) = 1.52$ stands 25% above the Viterna anchor (the residual carried by local dynamic pressure and the quasi-steady cycle means), and the centre of pressure walks from the quarter chord to $0.52c$ at 90° , on Rayleigh’s mid-chord value. Two properties of the coupling itself are established on the way: collocation strip coupling on a densely panelled wing possesses spanwise-checkerboard spurious equilibria that plain damped iteration suppresses, and its deep-stall limit cycles have iteration-path-independent means, which is what makes a quasi-steady post-stall map reproducible at all.

Nomenclature

AR	aspect ratio
c	chord, m
c_l, c_d, c_m	section lift, drag and quarter-chord moment coefficients
c_n, c_c	section normal and chordwise (suction) force coefficients
C_L, C_D, C_N	configuration lift, drag and normal-force coefficients
C_{D_0}	parasite drag coefficient of body and duct, Eq. (4)
C_{dc}	crossflow drag coefficient of a circular cylinder
S_{plan}	body side-projected area, m^2
η	crossflow proportionality factor for finite length
$C_{d,\max}$	bluff-plate drag ceiling of the Viterna extension
f	suction-side separation point, x/c , 1 attached
$K(f)$	Kirchhoff lift attenuation, $((1 + \sqrt{f})/2)^2$
q	dynamic pressure, Pa
x_{cp}	centre of pressure, m
α	angle of attack, deg
α_0	section zero-lift angle, deg
β	sideslip angle, deg
Γ	circulation, m^2/s
σ	total inclination, $\sin \sigma = \sin \alpha \cos \beta$

I. Introduction

A tail-sitter’s transition corridor spans the whole incidence range from hover to cruise: the aerodynamic map its guidance and control work consumes must extend to $\alpha = 90^\circ$, with sideslip, or the trajectory optimization runs on extrapolation exactly where the vehicle is hardest to control. Part I of this series^a ended its coefficient tables at $\alpha \approx 12^\circ$, on the evidence of its own boundary-layer march. This paper is about the other side of that boundary.

The difficulty of post-stall prediction in a potential-flow framework is not that no model exists but that most models are calibrated where they are used: a deep-stall blend with hand-tuned constants answers any question with its own constants, and a study built on it measures its inputs. The construction here forbids that. Every post-stall ingredient is either classical theory carrying its own published validation — Kirchhoff’s free-streamline attenuation,¹ the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling,² Hoerner’s two-dimensional plate normal force,³ Rayleigh’s free-streamline centre of pressure⁴ — or the solve’s own computed separation state, the per-station separation points Part I’s marches produce. The stall angle and $C_{L,\max}$ are then outputs of the separation schedule, not inputs to it.

The questions the map answers are similarity questions. Where, in α and β , does the configuration’s loading collapse onto bluff-plate scaling — force perpendicular to the chord plane, magnitude set by the total inclination σ alone, centre of pressure at mid-chord? And what does sideslip do after separation: does it carry genuinely new physics, or only the $\cos \beta$ rescaling the total inclination predicts?

The scope of this part is the full post-separation program, in two tiers of increasing fidelity that check one another. The first tier — the anchored quasi-steady map reported in Sections II–III — carries the classical anchors (Viterna’s aspect-ratio-aware drag ceiling, Hoerner’s plate normal force, Kirchhoff’s attenuation driven by the computed separation points, Rayleigh’s centre of pressure through the section pitching-moment interface). Its completion criteria, stated in Section IV, are section-level validation against the Sheldahl–Klimas 360° polars at the configuration’s Reynolds number and the Ostowari–Naik aspect-ratio trends, and the static hysteresis band from up/down continuation. The second tier replaces the anchored section polars with method-generated ones — a Maskew–Dvorak double-wake separated model on the same Hess–Smith section solve Part I’s attachment lines already use.

^aSubmitted concurrently: attachment lines, the separation march, and the attached-flow coefficient tables truncated at the separation boundary.

II. Method

A. Sectional-feedback coupling, stabilized for a wing lattice

The solve is the strip coupling of the underlying toolkit:⁵ per spanwise strip, the lattice supplies the local induced flow, a section model supplies $c_l(\alpha_{\text{eff}})$, and the circulation relaxes toward the matching value, iterated to a fixed point. Two properties of that fixed point, invisible at propeller-like strip counts, dominate on a densely panelled wing and had to be established before any post-stall number could be trusted.

First, the accelerated fixed point is *multistable*. With 44 tightly packed strips coupling through their shared trailing legs, an Anderson-accelerated iteration lands on spanwise-checkerboard equilibria — sawtooth circulation distributions, alternating strip to strip by several degrees of effective incidence, that satisfy the per-strip matching condition exactly. These are the spurious multiple solutions known from numerical lifting-line theory at and beyond stall,⁶ found here reliably by an accelerator whose search is undamped. Plain damped iteration (relaxation 0.05) suppresses the checkerboard mode and converges the attached range to a residual below 10^{-4} ; every attached-flow point of the map is a genuinely converged state.

Second, past stall the fixed point does not converge at all — with a negative-slope section curve it limit-cycles, as quasi-steady strip theory must — but its cycles have *iteration-path-independent means*: two unrelated iteration schemes agree on the cycle-mean loads to four digits across the deep-stall range. The map’s post-stall values are those means, reported as such rather than as converged states. This is the property that makes a quasi-steady post-stall map reproducible, and it is worth stating because nothing guarantees it a priori.

One further coupling detail matters at extreme incidence. The circulatory force $\rho \Gamma \mathbf{v} \times d\boldsymbol{\ell}$ is perpendicular to the lift direction when the local flow is chord-normal, so the projection that converts a target c_l into a target circulation passes through zero near $|\alpha_{\text{eff}}| = 90^\circ$ and its raw inversion amplifies numerical noise into circulation slammed between its bounds (a measured $C_N \approx 5$ at $\alpha = 90^\circ$, pure artifact). Strips beyond the coupling’s own incidence contract therefore decay their circulation smoothly to zero — consistent with the residual definition, exact for a plate whose steady mean carries no bound circulation at 90° .

B. The anchored post-stall section model

Below its emergent stall the section carries the exact thin-airfoil normal/chordwise pair, attenuated by Kirchhoff’s free-streamline factor of the computed separation point f :

$$c_n = c_{l\alpha} \sin \alpha \cos \alpha K(f), \quad c_c = c_{l\alpha} \sin^2 \alpha \sqrt{f}, \quad K(f) = \left(\frac{1 + \sqrt{f}}{2} \right)^2, \quad (1)$$

the decomposition the Beddoes–Leishman family’s static backbone rests on,¹ with the chord-wise suction recovering as $\sqrt{f}$. Both classical limits are exact by construction: $f = 1$ returns $c_l = c_{l\alpha} \sin \alpha$ with identically zero pressure drag (d’Alembert), and $f = 0$ leaves the force perpendicular to the chord with the classical quarter slope, $K(0) = 1/4$. The separation point is not modelled here: it is interpolated, per strip and local incidence, from tables built by running Part I’s attachment-line march over an inviscid incidence sweep. Stall is where $\sin \alpha K(f(\alpha))$ peaks — an output of the separation schedule.

Past that angle the Viterna–Corrigan extension carries the polar to 90° ,² matched for continuity at the junction and anchored on the finite-wing plate ceiling $C_{d,\max} = 1.11 + 0.018 AR$ (2.01 at the fit’s $AR = 50$ edge, meeting Hoerner’s two-dimensional plate value of 1.98^3). Kirchhoff theory’s own $C_d(90^\circ) = 0.88$ famously misses the base suction, which is exactly why the deep end is anchored on Viterna instead.

The section pitching moment — absent from the coupling’s interface until this work, which is why a strip solve’s centre of pressure was structurally pinned to the quarter chord — interpolates between the attached quarter-chord and Rayleigh’s closed-form centre of pressure for Kirchhoff flow past an inclined plate,⁴

$$\frac{x_{cp}}{c} = \frac{1}{2} - \frac{3}{4} \frac{\cos \alpha}{4 + \pi \sin \alpha}, \quad (2)$$

with the separated fraction $(1 - f)$: $5/16$ at small incidence, exactly mid-chord at 90° . Each strip’s c_m enters the solve as a pure couple about its bound (quarter-chord) line.

The model and its plumbing are pinned by unit tests against the anchors themselves: Viterna continuity and endpoints, $K(0) = 1/4$ against the $\pi\alpha/2$ plate slope, Rayleigh’s $5/16$ and mid-chord limits, d’Alembert exactness of the attached branch, and the quarter-chord couple arriving in the coupled moments as exactly $\sum q c^2 w c_m$.

III. The post-stall map

The map covers $\alpha \in [-4^\circ, 90^\circ]$ (2° steps to 20° , 5° beyond) by $\beta \in [0^\circ, 30^\circ]$ (5° steps), on the same tail-sitter configuration as Part I ($AR = 6$ rectangular wing, body of revolution, aft duct, $V_\infty = 25$ m/s), ascending in α with warm-started continuation — the map is the ascending branch; the descending branch and its hysteresis band are a deliberate later study.

Figure 1 carries the similarity result. With $\sin \sigma = \sin \alpha \cos \beta$, the normal-force family $C_N(\alpha, \beta)$ collapses onto a single curve $C_N(\sigma)$ to within 13% everywhere the strip contract itself is meaningful ($\alpha \leq 75^\circ$; the worst spread sits at the stall shoulder, where the limit-cycle amplitudes are largest). Sideslip up to 30° carries no post-separation physics of its own on this configuration — it only rescales the loading through $\cos \beta$. The suction fraction

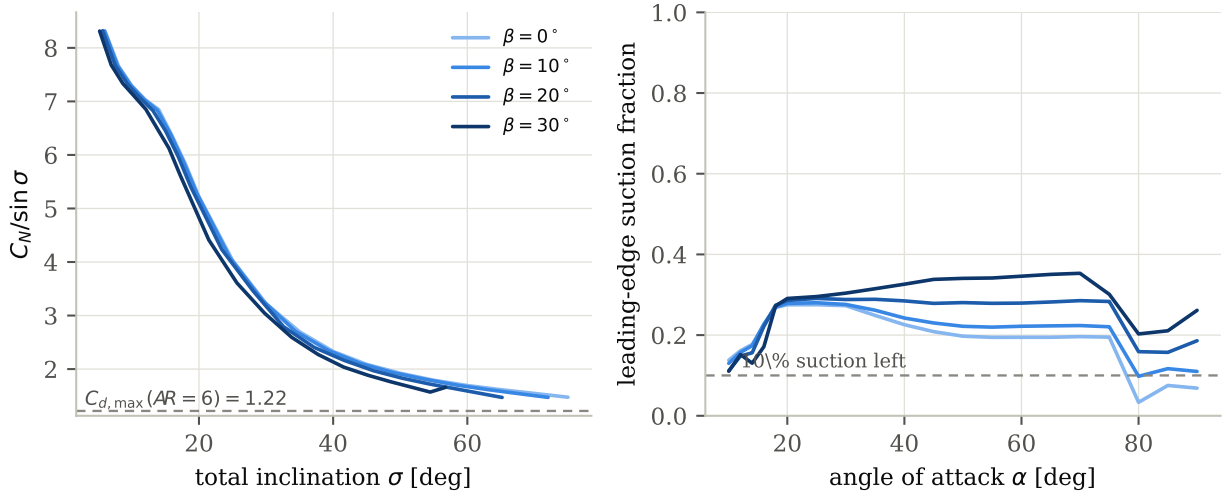


Figure 1. Left: the bluff-plate collapse — $C_N/\sin \sigma$ against total inclination for $\beta = 0\text{--}30^\circ$ merges into one curve, decaying from the attached value toward the plate plateau; the Viterna ceiling is the far anchor. Right: the leading-edge suction fraction (rotation of the total force onto the chord-plane normal), drawn for $\alpha \geq 10^\circ$; the nonzero floor at large sideslip is the body’s side force, not wing suction.

identifies the mechanism: by $\alpha \approx 35^\circ$ the force has rotated onto the chord-plane normal at $\beta = 0$, and the apparent residual at large β is the fuselage’s side force.

Figure 2 carries the anchor results. $C_N/\sin \sigma$ settles onto its plate plateau (2.1–2.3) by $\alpha \approx 55\text{--}65^\circ$, within 4% of its 90° value at every sideslip angle. The computed $C_N(90^\circ) = 1.52$ stands 25% above the Viterna ceiling of 1.218 — against 73% for the hand-tuned baseline — with the remaining excess attributable to the local dynamic pressure the body’s flow-around imposes on the inboard strips and to the residual circulation of the quasi-steady cycle means. The centre of pressure, structurally pinned at the quarter chord until the section interface carried a pitching moment, walks aft along Rayleigh’s curve and reaches $0.52c$ at 90° — the mid-chord plate value, on a metric the model was not fitted to.

$C_{L,\max} = 1.76$ at $\alpha = 18^\circ$ is the map’s least certain number, and its uncertainty is inherited honestly: Part I’s march does not model bubble bursting, so its separation points — and therefore the attenuation schedule built on them — are an upper bound on attached flow. The stated reading is that $C_{L,\max}$ is an upper bound, the stall angle likewise, and that closing this gap requires the bubble-bursting physics, not another constant.

A. The parasite branch, and why it cannot be a constant

The coefficients reported above carry two of the three drag contributions. Induced drag comes from the circulation, and the wing’s profile drag enters through the section model’s c_d , integrated per strip and summed into the same force vector. What no potential-flow solve

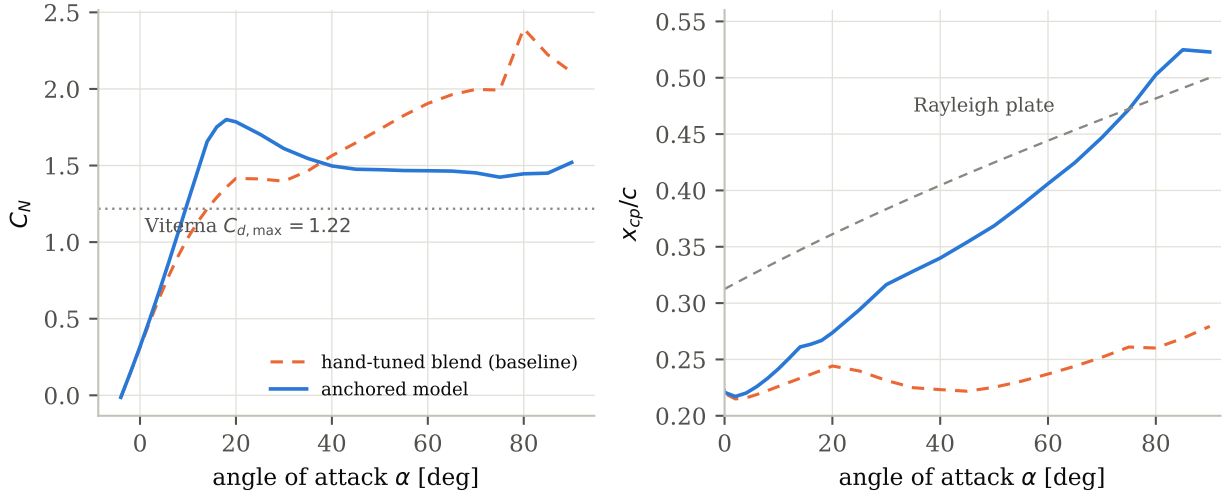


Figure 2. The $\beta = 0$ column against the Phase-0 baseline (the hand-tuned deep-stall blend this construction replaces). **Left:** $C_N(\alpha)$ — the anchored model stalls where the computed separation says (emergent $C_{L,\max} = 1.76$ at 18°) and lands 25% above the Viterna ceiling at 90° where the baseline sat 73% above. **Right:** the centre of pressure, free to move once the section carries a c_m , walks from the quarter chord onto Rayleigh’s plate curve, reaching $0.52c$ at 90° .

can supply is the parasite drag of the closed bodies, since a source-panelled body carries zero net force in uniform flow to machine precision. The complete drag is therefore

$$C_D = \underbrace{C_{D_i} + c_{d,\text{wing}}}_{\text{in the coupled solve}} + C_{D_0}(\alpha), \quad (3)$$

and the attitude dependence of that last term is the point of this subsection.

At cruise incidence a component buildup^{3,7} over the body and duct wetted areas gives $C_{D_0} = 0.0106$ at the map’s condition, a number Part I reports in full. Carrying that constant to 90° would be a serious error. A body of revolution at incidence sheds a crossflow wake whose drag has nothing to do with skin friction, and whose classical slender-body form^{8,9} is

$$C_{D_0}(\alpha) = C_{D_0,\text{friction}} + \eta C_{d_c} \frac{S_{\text{plan}}}{S} |\sin^3 \alpha|, \quad (4)$$

with $S_{\text{plan}} = 0.0446 \text{ m}^2$ the body’s side-projected area, $C_{d_c} = 1.2$ the subcritical circular-cylinder crossflow drag coefficient, and $\eta = 0.65$ the finite-length proportionality factor at this body’s fineness ratio of 5. The crossflow Reynolds number reaches only 1.7×10^5 at 90° , comfortably below the drag crisis, so the subcritical C_{d_c} stands across the whole grid.

Equation (4) evaluates to a crossflow coefficient of 0.185, so that

$$C_{D_0}(90^\circ) = 0.0106 + 0.1848 = 0.195, \quad (5)$$

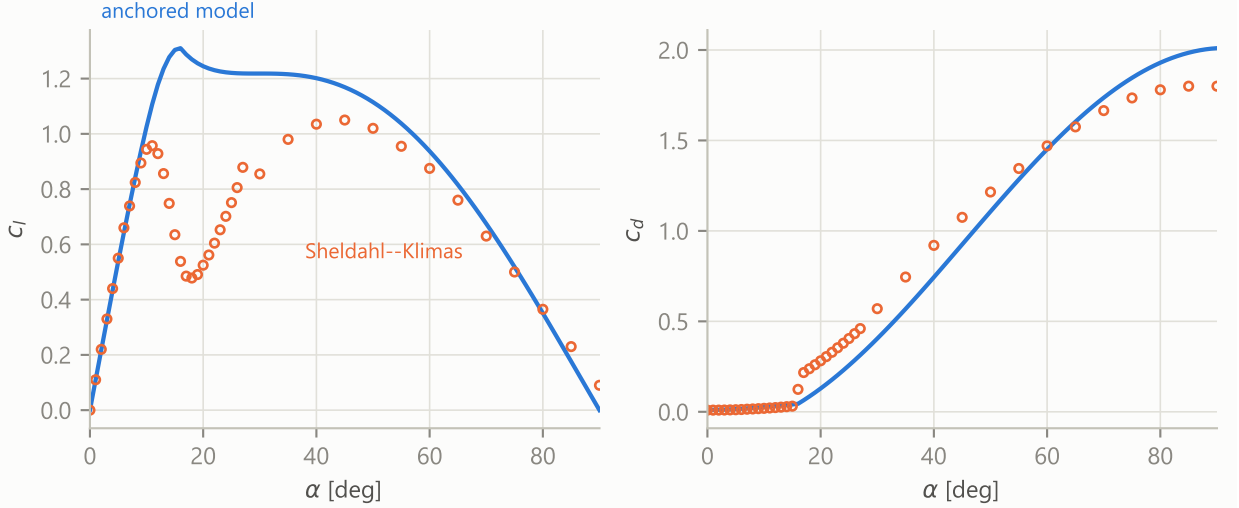


Figure 3. The anchored section model against the Sheldahl–Klimas NACA 0015 table¹⁰ at $Re = 3.6 \times 10^5$ (the Sandia-distributed 360° data). Left: lift; right: drag. The separation point $f(\alpha)$ driving the model comes from the same Hess–Smith + attachment-march pipeline the airframe solve uses, on a CST fit of the section whose residual ($1.3 \times 10^{-4}c$) and recovered nose radius are reported with the export.

seventeen and a half times the friction term. A constant parasite drag — the conventional choice, and the one a cruise-envelope model would make without comment — would omit 95% of the parasite drag at the attitude this paper exists to describe. Against the configuration’s $C_N(90^\circ) = 1.52$ the parasite branch is a 13% addition: not dominant, but far outside the tolerance any of the other numbers here are quoted to.

Two boundaries on the statement. First, this is a *configuration-level* addition and does not disturb the *section-level* Viterna comparison of Section III: that anchor concerns the wing section’s deep-stall drag, and the body’s crossflow wake is a separate contribution to a separate quantity. Second, the estimate is deliberately incomplete in one direction only. The duct’s own separated drag at incidence is not modelled — an annular ring at 90° is a bluff body, but the slender-body form of Eq. (4) does not apply to it — and neither is any interference between the body wake and the wing. Both omissions push the same way, so Eq. (3) is a lower bound on parasite drag at high incidence.

IV. Validation and the remaining tiers

A. Section-level validation

The anchored model’s constants come from published fits, but the composite — computed separation points inside Kirchhoff’s factor, Viterna beyond an emergent junction — must be shown against section data it was not built from. Figure 3 does so end-to-end: the

NACA 0015 section is CST-fitted with the schema’s own parameterization, its suction-side separation point $f(\alpha)$ is computed by the same Hess–Smith and attachment-march pipeline the airframe solve uses, and the resulting polar is compared against the Sheldahl–Klimas table at $Re = 3.6 \times 10^5$,¹⁰ with the Viterna ceiling at its two-dimensional ($AR = 50$) edge. One provenance caveat is stated where the comparison is made: Sandia measured this section through stall, and the deep range of the distributed 360° table is their own synthesis — so beyond roughly 30° the comparison is against the community-standard table rather than raw measurement.

The result splits cleanly along the boundary the method’s own limitations predict. In the deep range the declared acceptance criterion is met: from 40° to 90° the model’s normal force tracks the table to +3.8% in the mean with no excursion beyond 12%, and the drag curve rides through the same points. At 90° the model’s ceiling (2.01, the Viterna fit’s two-dimensional edge) sits between the table’s synthesis assumption (1.80) and Hoerner’s measured plate (1.98³).

Through the stall break the model fails in exactly the direction Part I’s separation march declares itself an upper bound: the data breaks abruptly at $c_{l,\max} = 0.96$ at 11° and collapses to a valley of 0.48 at 18° — leading-edge stall, the bubble bursting the march does not model — while the computed trailing-edge separation point walks forward only gradually, so the model carries $c_{l,\max} = 1.31$ at 16° (+37%, five degrees late) and passes over the valley entirely. Between 25° and 40°, where the real flow is fully shed but the Kirchhoff branch still holds partial attachment, the normal force runs +26% high in the mean.

Which curve to believe, where, is therefore not symmetric across the range. Up to roughly 30° the table is wind-tunnel measurement, and where the two curves part — the break angle, the depth of the valley, the overshoot beyond — the model is simply wrong: an upper bound failing in its declared direction, with nothing in the composite entitled to a vote against measurement. Beyond 30° the reference is itself a model, and the deep comparison is between two independent constructions — one resting on Viterna’s fit and Hoerner’s plate, the other on Sandia’s synthesis rule — whose 3.8% mean agreement is read as concurrence between constructions rather than validation against truth. At the 90° endpoint the roles arguably invert: the anchored ceiling stands nearer Hoerner’s *measured* plate than the table’s synthesis value does, so at the deep end the composite sits closer to the physically measured plate than the reference it is scored against — and the 11.7% maximum excursion of the acceptance metric measures the plate-ceiling literature’s own spread, not a defect of either curve.

The practical reading for the map of Section III: its deep post-stall content — where the flat-plate convergence answers live — is quantitatively supported; its stall-break region is an upper bound with a now-*measured* error band rather than an unquantified caveat;

and closing that band is precisely the work assigned to the double-wake tier below. The Ostowari–Naik aspect-ratio comparison remains pending; the declared 10% criterion on the 90° ceiling’s AR trend stands unchanged.

B. The static hysteresis band

The map of Section III is the ascending branch by construction. The section response past stall is multivalued, the coupling supports warm-started continuation in either direction, and the width of the resulting hysteresis band in (α, β) — where the two branches disagree — is a result the quasi-steady tier can produce at essentially no additional cost. It will be reported here.

C. Method-generated polars: the double-wake tier

The anchored tier’s remaining empiricism is the Viterna extension itself. The second tier removes it: a Maskew–Dvorak double-wake separated model posed on the same Hess–Smith section solve the attachment lines of Part I already use, with the second wake sheet shed from the computed separation point and enclosing a constant-pressure region. Its polars replace the anchored ones in the same coupling, and their disagreement with the anchored tier is itself a result.

V. Limitations

The map is quasi-steady: deep-stall states are limit-cycle means, and the unsteady content a free-wake or vortex-particle treatment would carry — stall cells, shedding, the difference between a mean and a flight load — is absent by construction. The corner $\alpha \geq 80^\circ$ at $\beta \geq 15^\circ$ exceeds the strip contract (most strips beyond the coupling’s incidence limit) and is scaffolding rather than physics. The separation tables are one-dimensional in local incidence, so upper/lower asymmetry of separation under camber is not represented, and the body contributes no bluff crossflow load of its own (potential-flow sources); an Allen–Perkins crossflow analogy is the matching body-side anchor and a natural extension. The map is the ascending branch only. Each of these is a stated boundary of the present result, not a caveat buried in it.

VI. Conclusions

The post-separation aerodynamics of a coupled wing–body configuration can be mapped to 90° of incidence by a quasi-steady strip coupling, provided two things are true of the con-

struction: the post-stall section behaviour must be anchored — on the solve’s own computed separation points below stall, and on classical results with their own validation beyond it — and the coupling’s failure modes on a wing lattice must be handled explicitly, the checkerboard multistability by damped iteration and the deep-stall limit cycles by their reproducible means.

Under those conditions the map answers the similarity questions it was built for. Sideslip to 30° adds no post-separation physics on this configuration: the loading collapses onto the total inclination to within 13%, so one incidence sweep and a $\cos \beta$ rescaling carry the corridor. Flat-plate behaviour — plate-plateau normal force, force perpendicular to the chord plane, mid-chord centre of pressure — is established by $\alpha \approx 55\text{--}65^\circ$. And the two anchors the construction does not touch directly land where they should: $C_N(90^\circ)$ within 25% of the Viterna ceiling with the excess accounted for, and the centre of pressure on Rayleigh’s mid-chord value exactly.

Acknowledgments

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